

Q40. (Theory) Explain QoS (Quality of Service) levels in MQTT: QoS 0 (At most once), QoS 1 (At least once), QoS 2 (Exactly once). In an IoT water level monitoring system, which QoS level would you choose and why?

Ans. In MQTT, QoS levels define the "contract" between the sender and the receiver regarding how a message is delivered.

QoS 0 (At most once): This is the fastest method but the least reliable. The sender fires the message once and does not care if it is received or lost. It is like sending a postcard; if it gets lost in the mail, you will never know.

QoS 1 (At least once): This ensures the message arrives at least one time at the receiver. The sender keeps trying until it gets an acknowledgment ("ACK") back from the receiver. However, there is a small chance of getting duplicate messages if the acknowledgment gets lost.

QoS 2 (Exactly once): This is the most reliable (and slowest) method. It uses a four-step handshake process to guarantee that the message is received exactly one time, with no duplicates and no loss.

Choosing for an IoT Water Level Monitoring System

For an IoT water level monitoring system, the best choice depends on the specific requirement, but QoS 1 is generally recommended.

Why QoS 1?

In a water monitoring system, it is vital that you receive the data (e.g., "Water tank is full") to prevent overflow. QoS 0 is too risky because losing a critical alert could lead to a disaster.

Why not QoS 2?

QoS 2 requires much more back-and-forth communication, which consumes more battery and bandwidth. Since a water level monitoring system usually just needs to know the state of the tank, the slight risk of a duplicate message (which the system can easily ignore) is a better trade-off than the high power usage of QoS 2.